

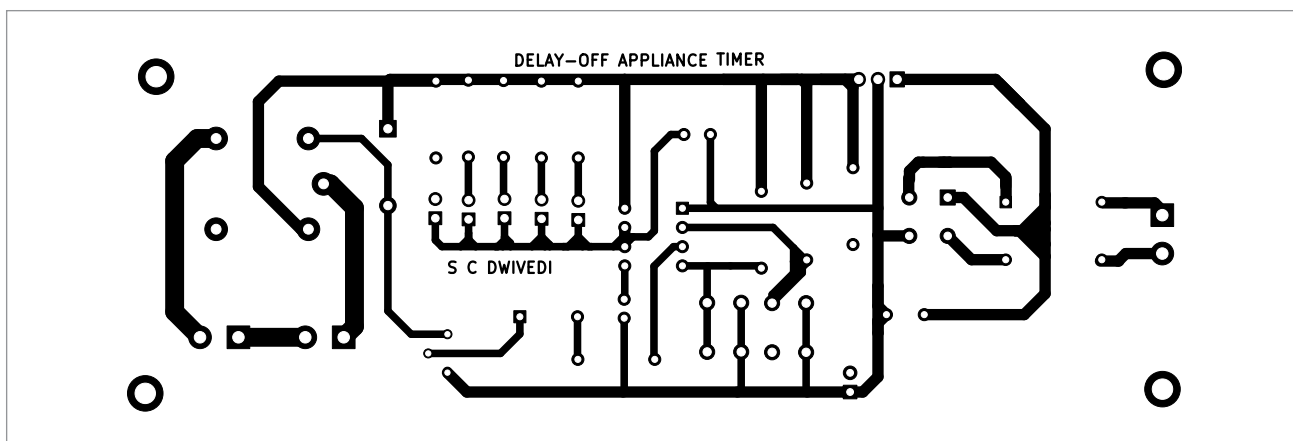


Fig. 3: Actual-size PCB layout

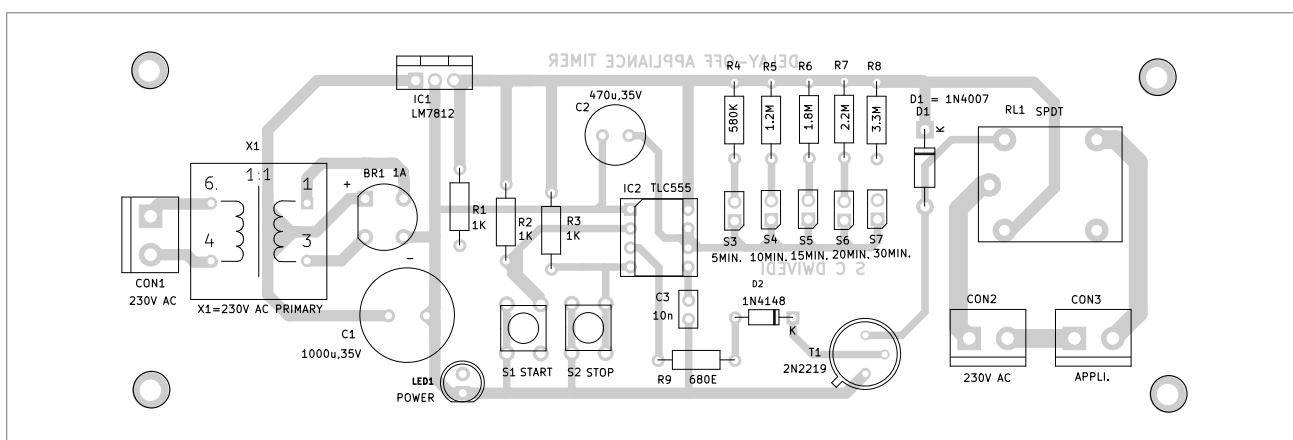


Fig. 4: Component side

A power LED (LED1) indicates the supply status. If LED1 is lit, the power is on; if LED1 is unlit, the power is off.

The TLC555 timer (IC2), the heart of the circuit, is configured in monostable mode. The timing capacitor C2 (470 μ F, 35V) and one of the resistors R4 through R8 (selectable via switches S3 through S7) determines the off-delay time. Momentarily pressing the start switch (S1) triggers IC2, causing its output to go high for the preset duration. During this time, the NPN transistor 2N2219 (T1) conducts and energises the relay (RL1). The connected appliance across CON3 receives power through the relay contacts, provided that CON2 is connected to a 230V AC source.

Once the set time elapses, the output of IC2 goes low, cutting off

transistor T1 and de-energising the relay, thereby switching off the appliance. The stop switch (S2) can be used to stop the appliance at any time during the countdown immediately.

The circuit is designed with five preset time-delay options, selectable via push-button switches. Switch S3 sets a 5-minute delay, switch S4 for 10 minutes, switch S5 for 15 minutes, switch S6 for 20 minutes, and switch S7 for 30 minutes. These switches determine the resistance in the RC timing network, thereby setting the desired delay duration.

In addition to the core components, several supporting elements ensure reliable operation. Diode D1, connected across the relay coil, acts as a flyback diode to clamp high-voltage spikes and protect transistor T1 and nearby components. Diode D2 blocks reverse voltage to

protect the base-emitter junction of T1, limits the base current for safe operation, and enhances switching reliability. Resistor R1 serves as a current-limiting resistor for LED1, while capacitor C3 (10nF) functions as a noise filter on the control pin of the TLC555 timer, thereby improving stability and preventing false triggering.

Construction and testing

An actual-size, single-sided PCB layout for the delay-off appliance timer is shown in Fig. 3, with the component layout in Fig. 4. After assembling the components on the PCB, the circuit should be enclosed in a suitable cabinet, ensuring proper openings are provided at the rear for mounting and ventilation. The switches (S1 through S7) should be mounted on the front